

COLOR AMP TAPE Technical Manual

Saturation, amplifier, tape, and harmonic color

Add character through distinct amp, tube, tape, and body-focused engines.

Included products: APIS, PtahTube, KHEPRI, SOVEREIGN

Routing authority: Music/color route: KHEPRI -> PTAHTUBE -> APIS. SOVEREIGN is a source-oriented amp alternative.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

APIS

Body and color processor built around color, body, petal dispersion, air, protected I/O, and channel-mode support.

Declared product: \${PROJECT_NAME} | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
INPUT	Input	Float
COLOR	Color	Float
BODY	Body	Float
FLOWER	Flower Petal	Float
AIR	Air	Float
OUTPUT	Output	Float
channelmode	Channel Mode	Float
dualmonolink	Dual Mono Link	Float
gainmatch	Gain Match	Bool

Artwork authority

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PtahTube

Phi-based tube color processor with input, drive, energy, balance, focus, output, and Flower of Life dispersal tied to its illuminated tube field.

Declared product: PTAHTUBE | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
input	Input	Float
drive	Drive	Float
output	Output	Float
energy	Energy	Float
balance	Balance	Float
power	Power	Float
focus	Focus	Float
flower	Flower of Life Dispersal	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

Assets/Ptah.png

Factory programs

MANUAL | USER SETTINGS

KHEPRI

Sacred tape machine with selectable machine, Phi formula, tape type (Gold Alloy, Ferric Oxide, Metal Alloy), speed, drive, bias, wow, flutter, head bump, HF rolloff, and parallel mix.

Declared product: Khepri | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
machine	Machine	Choice
formula	Phi Formula	Choice
tapeType	Tape Type	Choice
speed	Speed	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
input	Input	Float
drive	Drive	Float
bias	Bias	Float
wow	Wow	Float
flutter	Flutter	Float
bump	Head Bump	Float
rolloff	HF Rolloff	Float
mix	Mix	Float
output	Output	Float

Artwork authority

Assets/Khepri_Background.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT VU	Ballistic input level	Calibrated to the printed analog VU scale.
OUTPUT VU	Ballistic post-tape level	Compare with INPUT to avoid loudness bias.
Waveform	Recent live output samples	Shows real signal motion, not a decorative animation.
Reels / beetle glow	Tape motion and drive state	Operational feedback; not a calibrated level meter.

Metal Alloy's high-frequency ceiling is constrained to 45 percent of the active sample rate before coefficient generation, keeping the filter below Nyquist at every supported host rate. A final finite-sample containment guard resets local tape memory and suppresses an invalid sample if upstream state ever becomes non-finite.

Factory coverage now spans all three tape stocks: Gold Alloy for balanced reference and glue, Ferric Oxide for vocal/body/vintage uses, and Metal Alloy for clean print, drum transient, and air uses. The former Full Reel and Low Reel extremes are not part of the current factory program table.

SOVEREIGN

Source-oriented amplifier and tone stage with drive, bass, mid, treble, presence, and master controls.

Declared product: \${PROJECT_NAME} | **Version:** 3.15 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Mix Bus | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
drive	Drive	Float
bass	Bass	Float
mid	Mid	Float
treble	Treble	Float
presence	Presence	Float
master	Master	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Artwork authority

source/Sovereign.png